

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

Due to the sudden web server page being unresponsive, it was caused by a large number of TCP SYN requests from unknown IP addresses, which is why there is not a proper connection being established.

According to Wireshark TCP/HTTP log, there seems to be one source IP address **"203.0.113.0"** coming out of nowhere and does not correspond to the company web server.

Direct Denial of Service SYN attack due to a single source of IP address, and causing unlimited SYN packets to flood the network traffic.

Section 2: Explain how the attack is causing the website to malfunction

1. first step is SYN packet: sending the first synchronized request when the client is trying to connect to a company webserver.

2. SYN ACK packet: this is making sure the first SYN/synchronise request has reached the device.

3. ACK packet, to acknowledge and confirm that connection is established, in TCP protocol.

Once a malicious actor sends a large number of SYN packets all at once, all the network traffic will be interrupted by flood and logs that keep track of the network, it will delay and take forever to form a proper connection. Therefore, the log shows "Gateway Time-out (text/html) error message." which means the web server takes longer than usual, and the gateway server will eventually stop responding, and send a timeout error.

Inside the log, the first column tells us the log number which keeps track of the network attempting to establish, and then it shows the source and destination IP address, where the packets get travelled. Then the info column shows that the network has attempted to create three-way handshake where it starts with SYN request, SYN ACK request, and then ACK confirming that connection has established.

And an employee visitor with the IP address of 198.51.100.0/24 are coming from employee computers and "203.0.113.0" is coming from attacker's IP address